

NAME SCHOOL TEACHER 

Pre-Junior Certificate Examination, 2017

Mathematics

Paper 1

Higher Level

Time: 2 hours, 30 minutes

300 marks

School stamp

Running total	
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For examiner			
Question	Mark	Question	Mark
1		11	
2		12	
3			
4			
5			
6			
7			
8			
9			
10		Total	

Grade

Instructions

There are 12 questions on this examination paper. Answer **all** questions.

Questions do not necessarily carry equal marks. To help you manage your time during this examination, a maximum time for each question is suggested. If you remain within these times you should have about 10 minutes left to review your work.

Write your answers in the spaces provided in this booklet. You may lose marks if you do not do so. There is space for extra work at the back of the booklet. You may also ask the superintendent for more paper. Label any extra work clearly with the question number and part.

The superintendent will give you a copy of the *Formulae and Tables* booklet. You must return it at the end of the examination. You are not allowed to bring your own copy into the examination.

You will lose marks if you do not show all necessary work.

You may lose marks if you do not include the appropriate units of measurement, where relevant.

You may lose marks if you do not give your answers in simplest form, where relevant.

Write the make and model of your calculator(s) here:

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Question 1

(Suggested maximum time: 10 minutes)

- (a)** Write down the fraction that is half-way between $\frac{1}{2}$ and $\frac{2}{3}$.

[illegible]

- (b)** Write the following four numbers in order, from the smallest to the biggest.

$$\frac{1}{8}, \quad 1.3 \times 10^{-1}, \quad 0.14, \quad 11.5\%.$$

[illegible]

- (c) Mount Everest is the highest mountain peak on Earth, located in the Mahalangur mountain range in Nepal and Tibet. Its summit is 8800 metres above sea level, correct to the nearest 100 metres.



Complete the inequality below, where h is the height of Mount Everest.

$\boxed{} \leq h < \boxed{}$

- (d)** On Budget Day in October 2016, Ireland's National Debt had fallen to €184 932 672 554.



- (i)** Express this figure correct to six significant figures.

[illegible]

- (ii) At the peak of the recession in 2012, Ireland's National Debt was €215 539 million. Using your answer to part (i), find the **percentage decrease** in our National Debt. Give your answer correct to one decimal place.

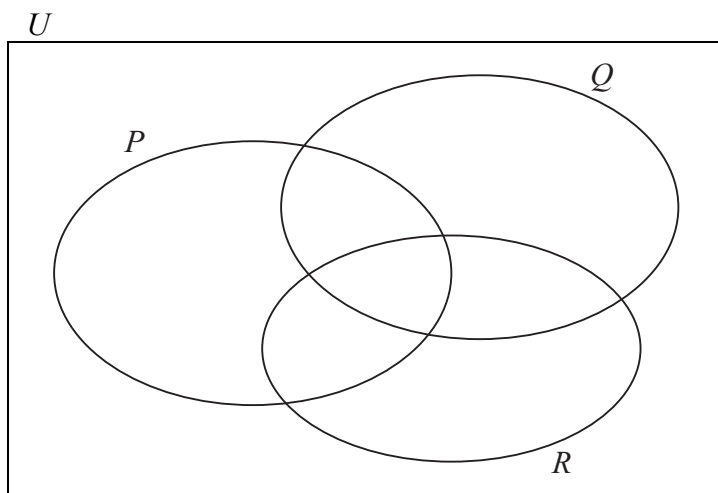
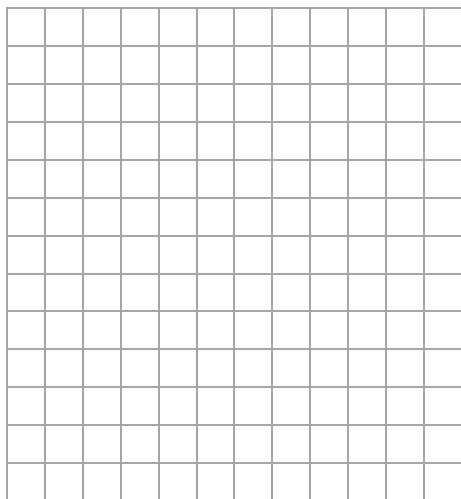
[illegible]

Question 2

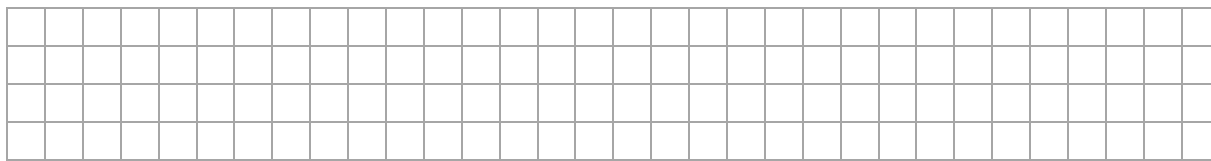
(Suggested maximum time: 10 minutes)

- (a) U is the set of natural numbers less than 20.
 P is the set of divisors of 15.
 Q is the set of divisors of 18.
 R is the set of prime numbers less than 20.

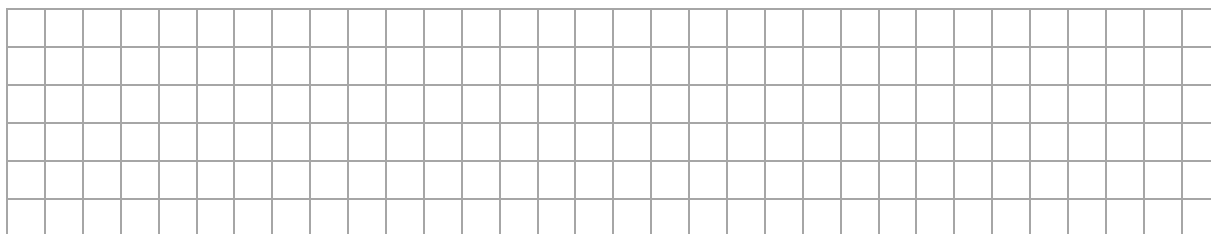
(i) Use this information to fill in the Venn diagram below.



(ii) Hence, or otherwise, write down the highest common factor of 15 and 18.

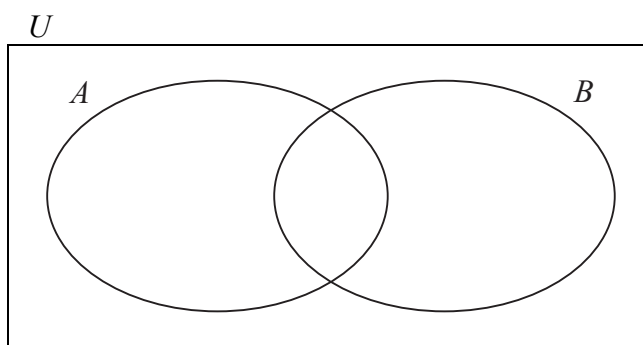
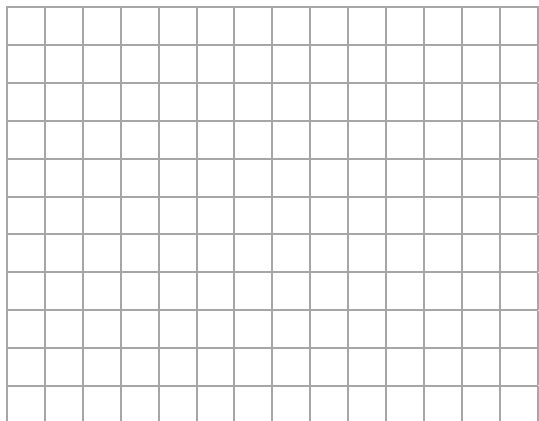


(iii) Calculate $\#(P \cup Q \cup R)'$.

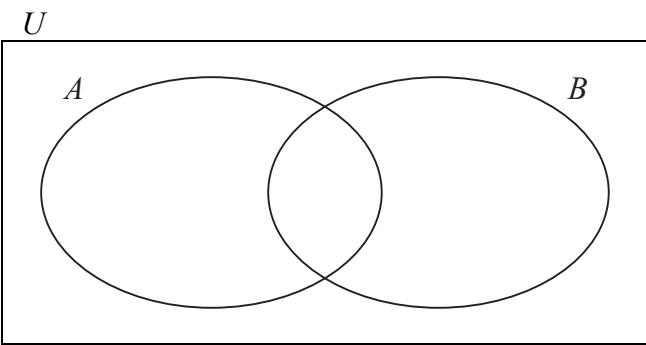
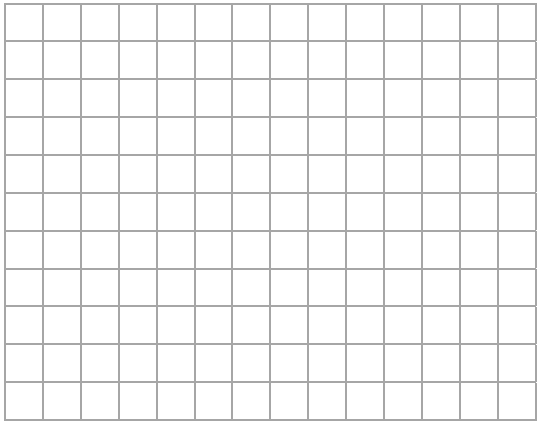


- (b) U is the universal set. A and B are two subsets of U .
 $\#U = 28$, $\#A = 18$ and $\#B = 8$.

(i) Find, with the aid of the Venn diagram, the minimum value of $\#(A \cup B)'$.



(ii) Find, with the aid of the Venn diagram, the maximum value of $\#(A \cup B)'$.



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Question 3

(Suggested maximum time: 10 minutes)

- (a) A supermarket sells a single 32.5 g packet of cheese and onion flavoured crisps for €0.545, excluding VAT.
If the current rate of VAT on crisps is 23%, calculate the price of a packet of crisps including VAT. Give your answer correct to the nearest cent.

[illegible]

- (b)** The same supermarket is also offering the following multipack deals:

Special Offer 1

6-pack (6 × 25 g)
of cheese and onion
flavoured crisps
only €1.99

Special Offer 2

14-pack (14 x 25 g)
of variety
flavoured crisps
only €3.99

- (i)** Which special offer is better value?
Give a reason for your answer.

[illegible]

- (ii) Paula says “The price per gram when purchasing a 6-pack of cheese and onion crisps is almost 50% cheaper than the price per gram when purchasing a single packet.” Is she correct? Give a reason for your answer.

Answer:

Reason:

- (c) Give two limitations of special offers in general.

[illegible]

Question 4

(Suggested maximum time: 10 minutes)

- (a)** For each of the following, write down the value of x , where $21 \leq x \leq 27$ and $x \in \mathbb{N}$.

Description	Value of x
A square number	
A multiple of 7	
A factor of 66	
A cube number	
A prime number	

[illegible]

- (b)** $\sqrt{27} + \sqrt{75} - \sqrt{12} = k\sqrt{3}$, where $k \in \mathbb{Z}$.
Find the value of k .

[illegible]

- (c) (i)** Simplify $(\sqrt[3]{x})^6$.

[illegible]

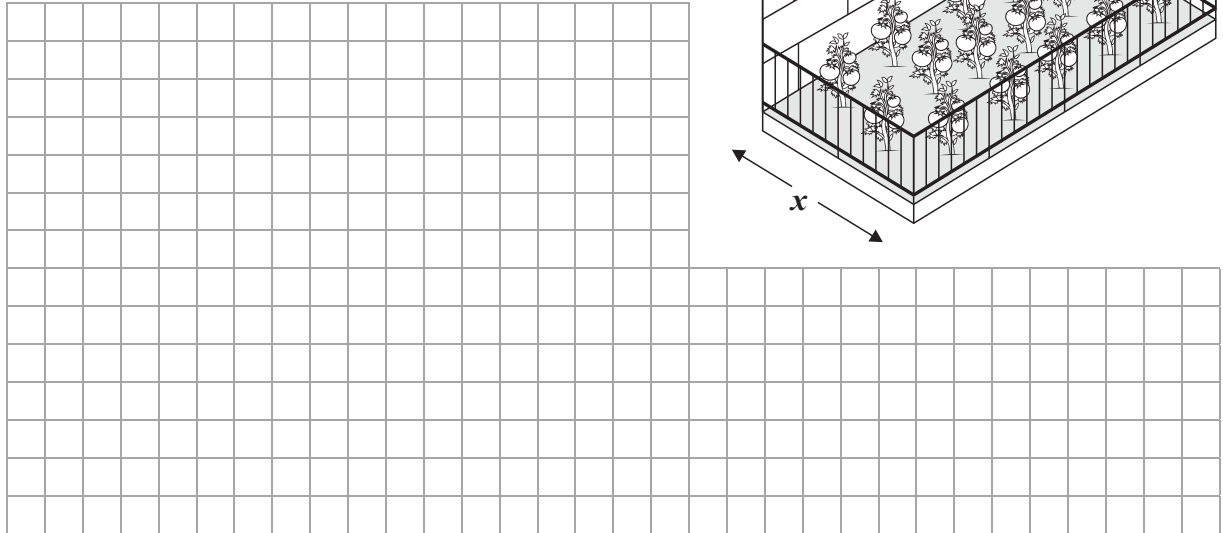
- (ii) Simplify $\frac{3y^2 \times (10y)^3}{2y^5}$.

[illegible]

Question 5**(Suggested maximum time: 15 minutes)**

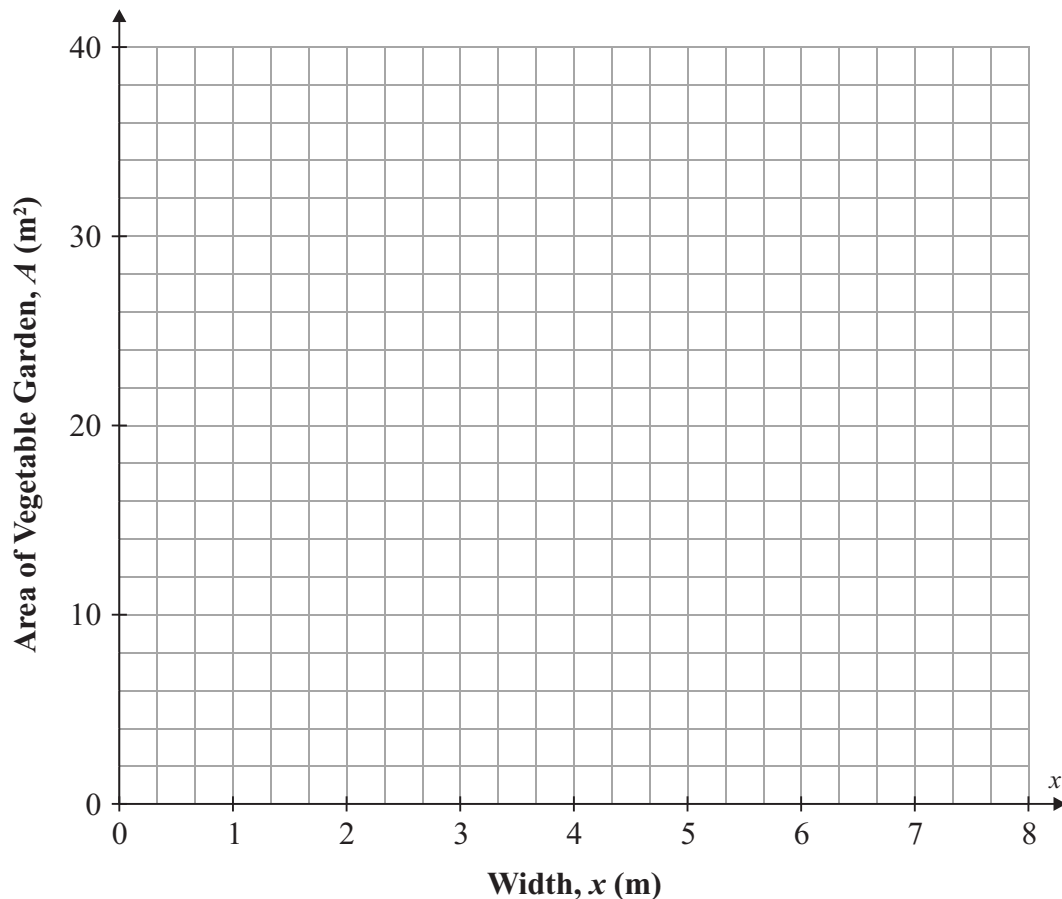
A school has a rectangular vegetable garden. The wall of the school building forms one side of the garden and a fence, of length 16 m, is used to form the remaining three sides, as shown. The width of the garden is x m.

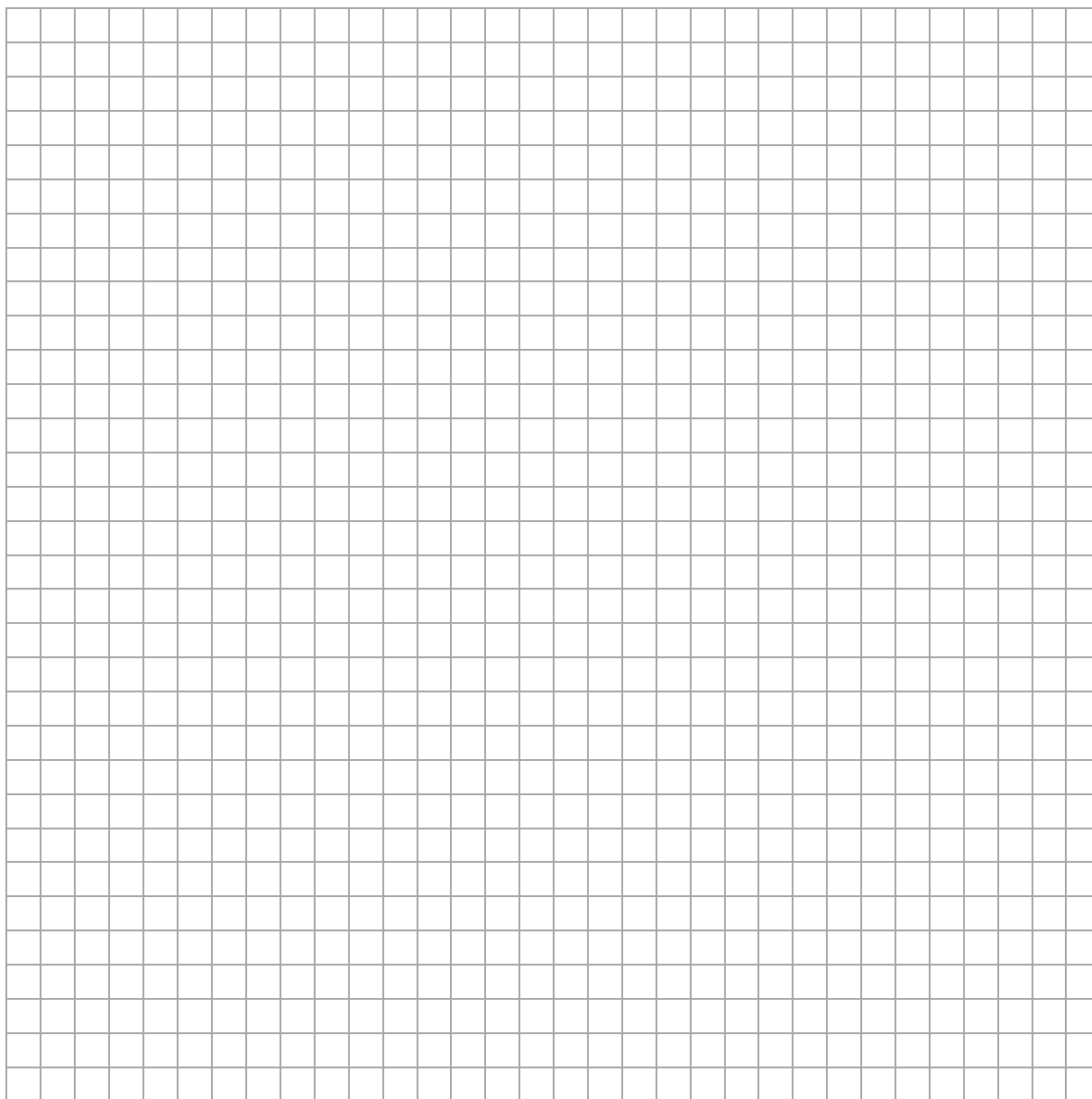
- (a) Show that the enclosed area of the garden, A m², is given by the function $A(x) = 16x - 2x^2$.



- (b) On the axes below, draw the graph of the function $A(x) = 16x - 2x^2$ for $0 \leq x \leq 8$, where $x \in \mathbb{R}$.

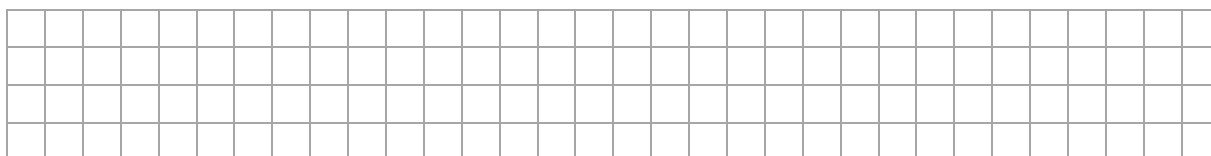
There is room for working out on the next page.



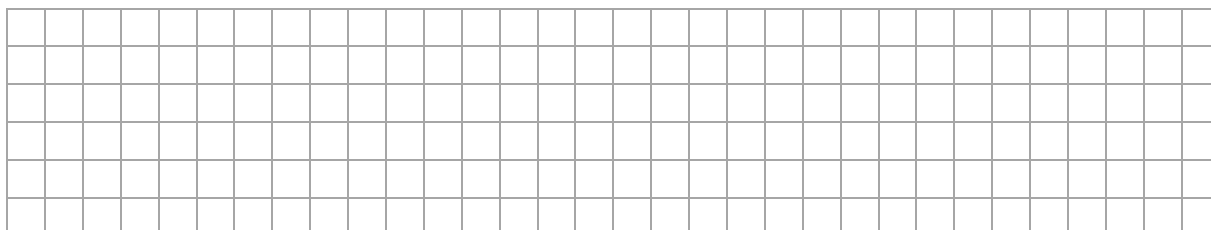


(c) Use your graph in part (b) to answer the following questions.

(i) Estimate the maximum possible area of the garden.



(ii) Find the range of values of x for which the area of the garden is greater than 26 m^2 .



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Question 6

(Suggested maximum time: 15 minutes)

(a) $p = 2a + c$.

(i) Write a in terms of p and c .

[illegible]

(ii) Hence, find the value of a when $p = 5$ and $c = \frac{1}{5}$.

[illegible]

(b) Solve the equation $x^2 - 6x + 4 = 0$.
Give each answer in surd form.

A full-page sheet of graph paper featuring a uniform grid of thin, light gray lines forming small squares across the entire surface. The grid consists of approximately 20 columns and 20 rows of squares.

(c) Write the following as a single fraction in its simplest form.

$$\frac{2x-3}{2} - \frac{4x-7}{5}.$$



Question 7

(Suggested maximum time: 10 minutes)

Aoife is a sales representative for a pharmaceutical company. She earns a basic pay of €2750 per month plus commission of 1.5% on her total sales.

For the month of December, Aoife's gross pay, including commission, was €4700.



- (a) (i)** Calculate Aoife's commission for the month of December.

[illegible]

- (ii)** Hence, calculate Aoife's total sales for the month of December.

[illegible]

The standard rate of tax is 20% and the higher rate of tax is 40%.

The standard rate cut-off point is €33 800 per annum.

- (b)** Aoife pays €1026.34 in income tax for December. Calculate Aoife's tax credits for the month.

[illegible]

Question 8

(Suggested maximum time: 10 minutes)

- (a)** The n th term of a sequence is given by the formula

$$T_n = 5n^2.$$

- (i)** Write down the first three terms of the sequence.

A large grid of graph paper with 20 columns and 10 rows. The grid is composed of small squares, with a slightly larger square at the top left corner, likely for a title or header. The grid is empty and ready for use.

- (ii)** Is 1000 a term in this sequence?
Justify your answer.

Answer:	
Justification:	

- (b)** Stephen plays golf.

The table below shows the average distance, in metres, he regularly hits the ball with various golf clubs.

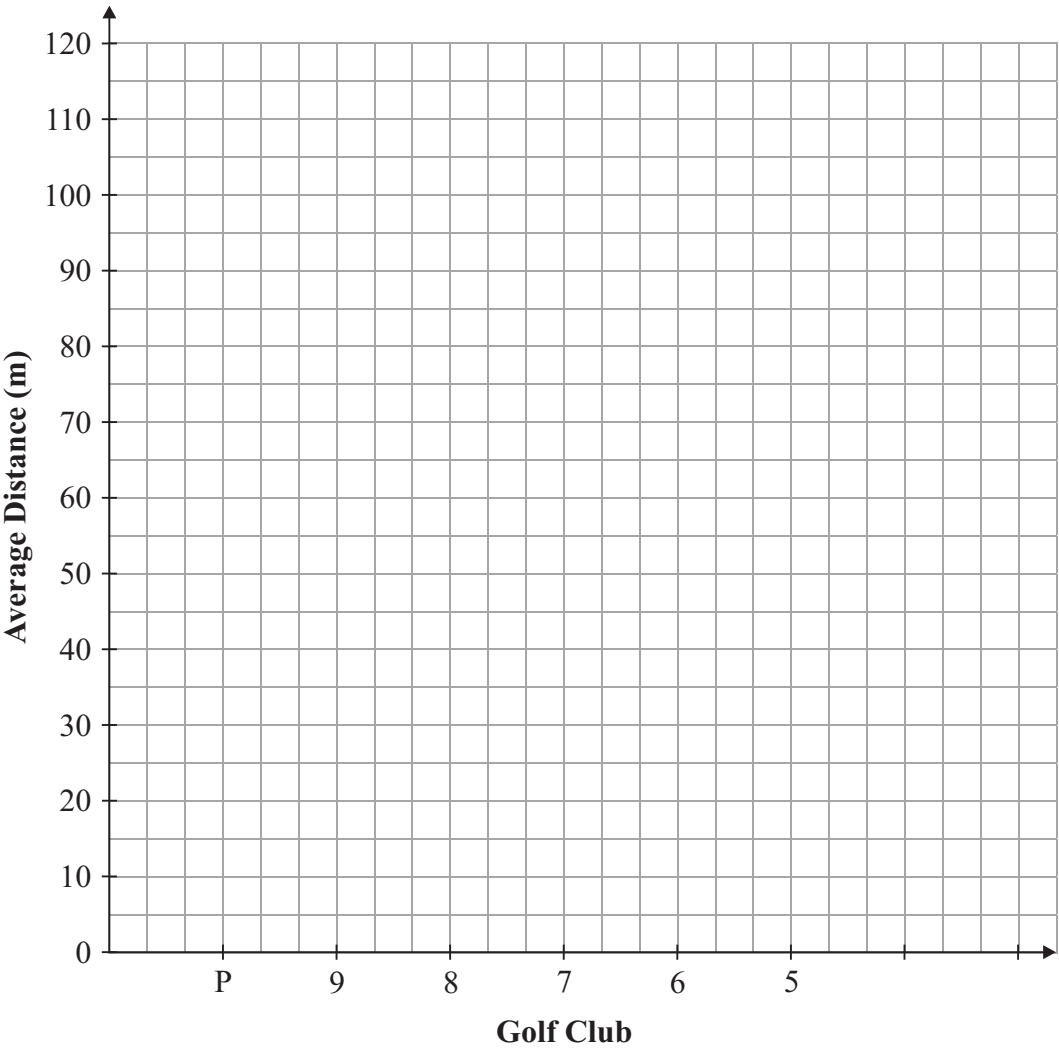
Golf Club	Putter	9 Iron	8 Iron	7 Iron	6 Iron	5 Iron
Average Distance (m)	75	80	85	90	95	100



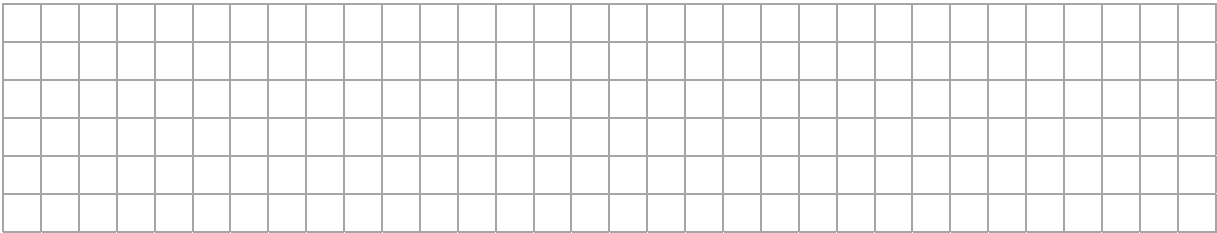
- (i) Is the pattern of distances shown in the table linear, quadratic or exponential? Justify your answer.

Answer:	
Justification:	

- (ii) On the grid below, draw a graph showing the pattern of average distances for the different golf clubs.



- (iii) Using your graph, or otherwise, estimate the average distance Stephen would expect to achieve with a 3 Iron golf club.

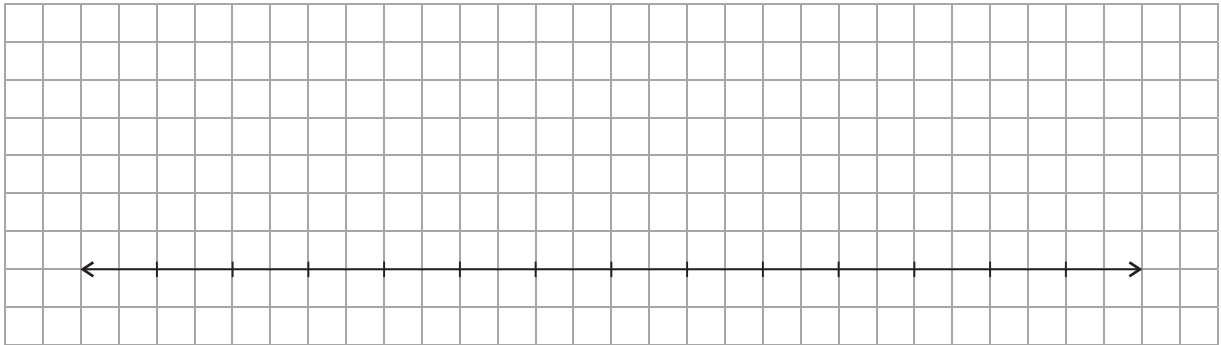


Question 9

(Suggested maximum time: 15 minutes)

- (a) Solve the following inequality and graph your solution on the number line.

$$-21 \leq 3 - 6x < 12, x \in \mathbb{Z}.$$



- (b) Factorise fully each of the following expressions.

(i) $27x^2 - 45x$

(ii) $6xy - 3sy - 4tx + 2st$

- (c) A 99 Flake ice cream, more commonly known as a **99** or **ninety-nine**, is an ice cream cone in which a Flake bar is inserted.

Mary buys 2 plain ice cream cones and 3 ninety-nines costing €10.20.
Joe buys 3 plain ice cream cones and 4 ninety-nines costing €14.20.
Calculate the cost of a flake.



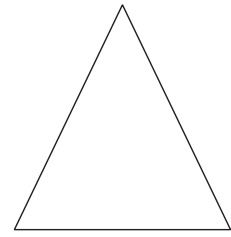
Question 10

(Suggested maximum time: 10 minutes)

A triangle has three sides as follows

$$3x + 2, \quad 2x - 5, \quad x + 1$$

where $x \in \mathbb{R}$.



- (a) Calculate the length of the perimeter of the triangle in terms of x . Give your answer in its simplest form.

- (b)** Show that the triangle is isosceles when x is equal to 6.

[illegible]

- (c) Mark says that x equal to 6 is the only value of x such that the triangle is isosceles. Is he correct? Justify your answer.

Answer:	
Justification:	

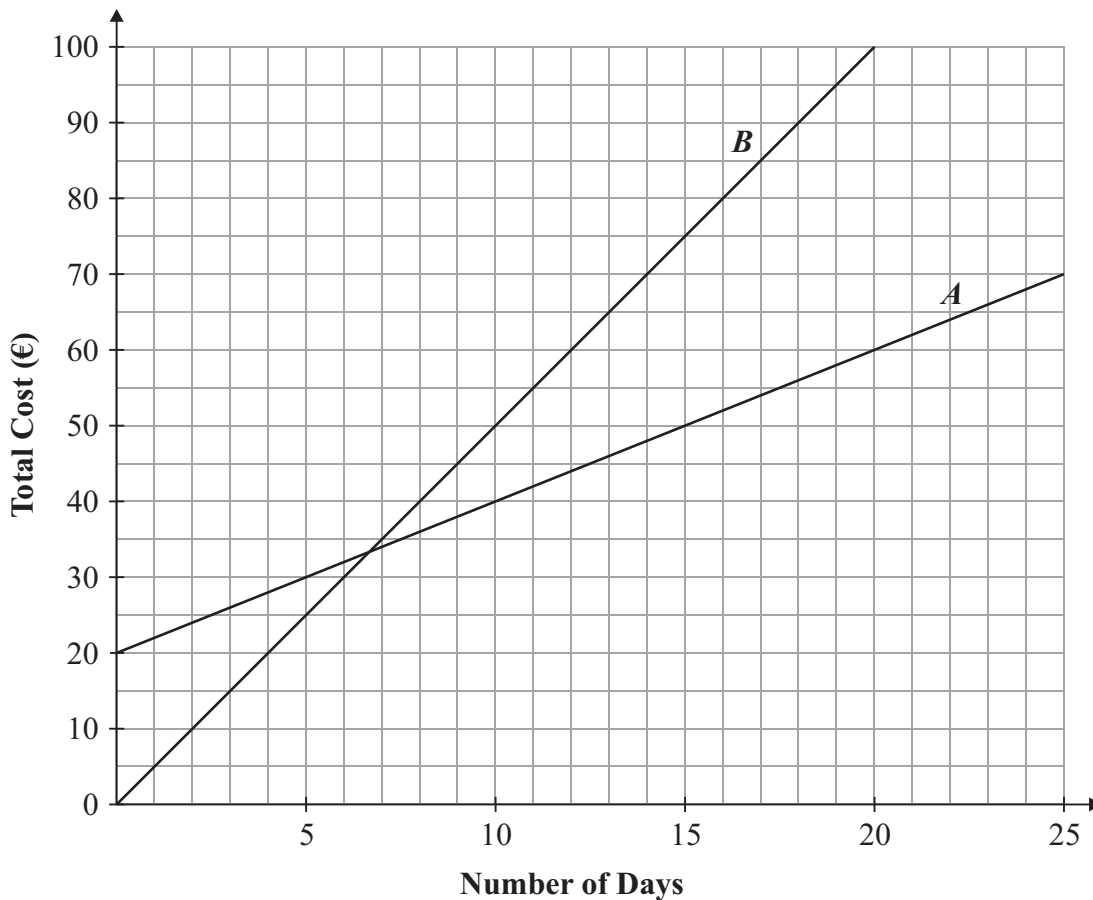
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Question 11

(Suggested maximum time: 15 minutes)

Two gyms, A and B , offer special rates to students for the month of July.

The grid below shows the graphs of the total cost of membership for the number of days used, for each gym, during July.



- (a)** Which gym has a joining fee?
Justify your answer with reference to the graph above.

[illegible]

- (b)** How might a student decide which gym is better value for money? Justify your answer.

Answer:	
Justification:	

- (c) Find the slope (rate of change) of the graph for gym *A*. Explain what this value means. Refer to both the cost and the number of days the gym is used in July.

Slope: _____

Explanation: _____

- (d) Write down a formula to represent the total cost for the number of days gym A is used in July. State clearly the meaning of any letters you use in your formula.

[illegible]

- (e) Write down a formula to represent the total cost for the number of days gym B is used in July. State clearly the meaning of any letters you use in your formula.

[illegible]

- (f)** Use your formulae from parts **(d)** and **(e)** to justify your answer in part **(b)**.

[illegible]

Question 12

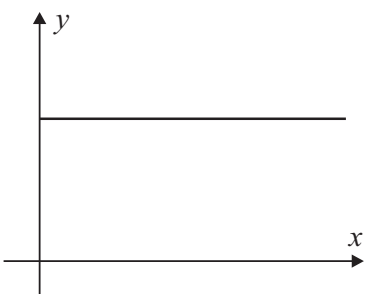
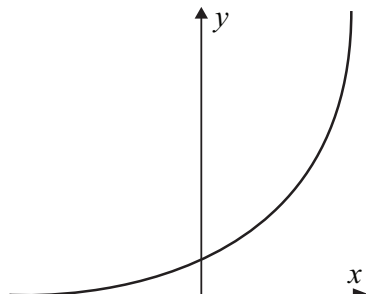
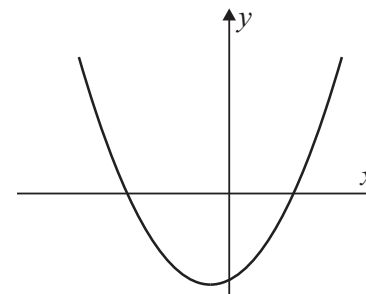
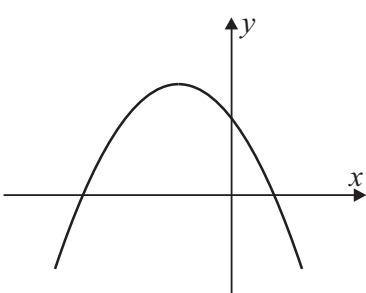
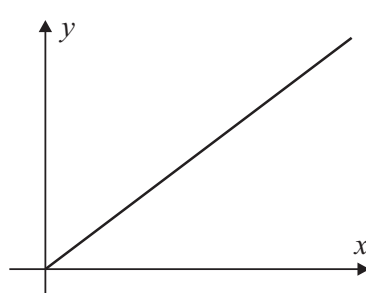
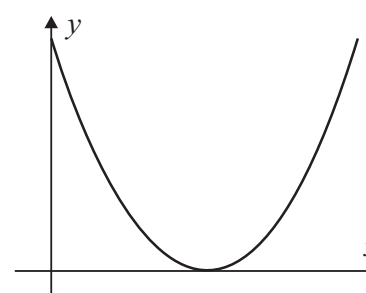
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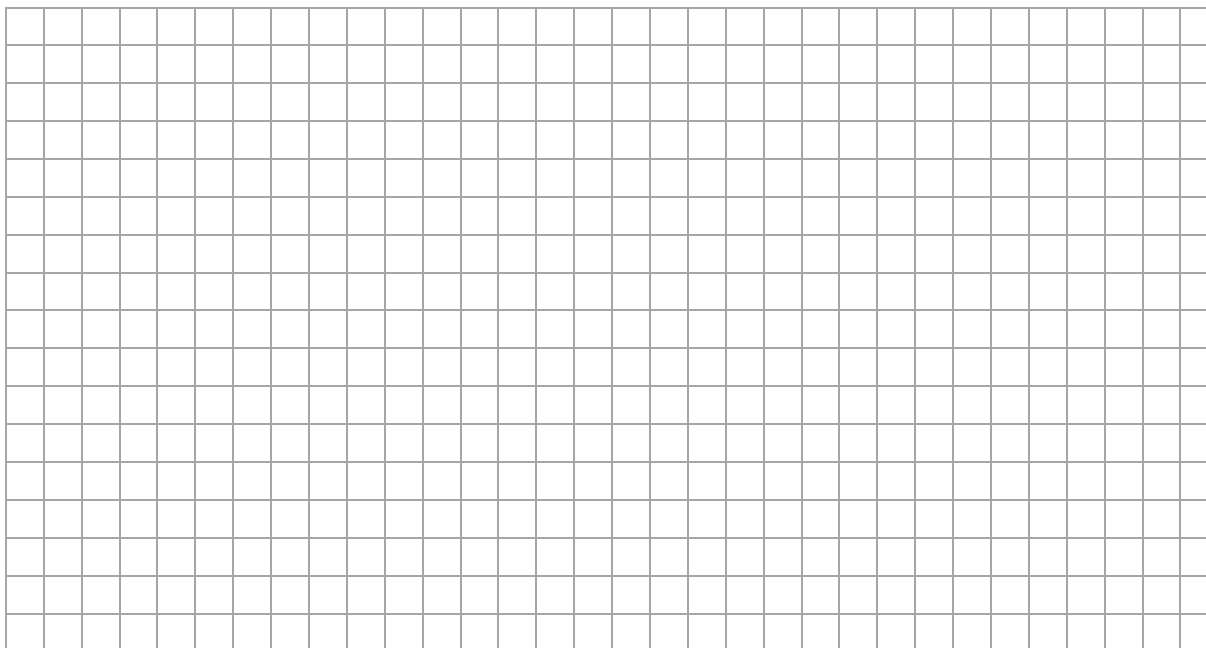
(a) Six functions are defined as follows:

$e(x) = 4 - 3x - x^2$	$f(x) = 4$	$g(x) = (x - 2)^2$
$h(x) = x^2 + x - 6$	$j(x) = 2^x$	$k(x) = \frac{4}{5}x$

The table below shows the sketches of all six functions.

Write the name of the function into the box underneath the correct sketch for each of the functions.



(b) Let f be the function $f: x \mapsto 2x^2 + 9$, where $x \in \mathbb{R}$.

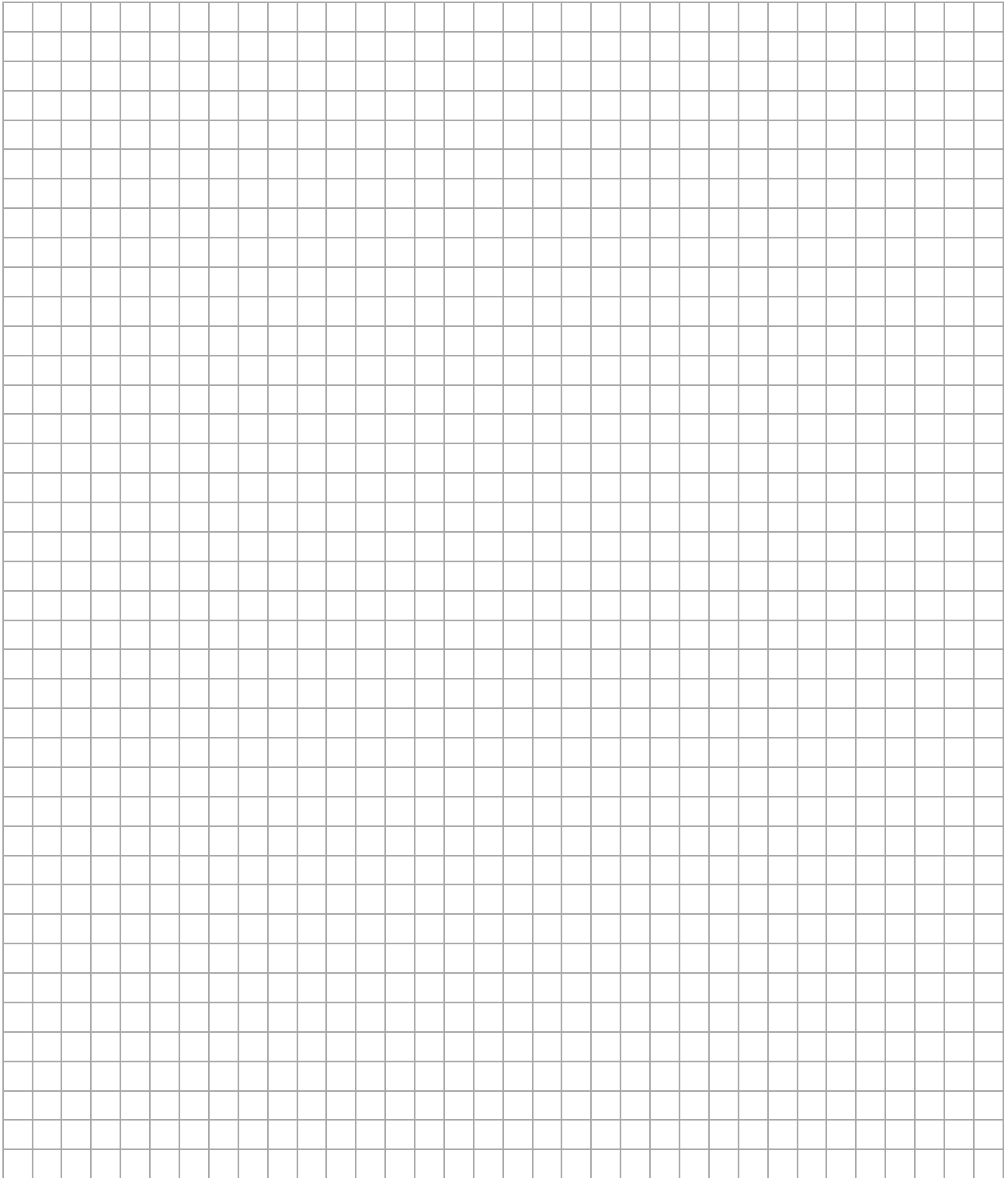
(i) Find the values of x for which $f(x) = 9x$.

A full-page sheet of white graph paper with a light gray grid. The grid consists of small squares, approximately 1 cm by 1 cm each. There are 20 columns and 15 rows of squares. The grid covers most of the page, leaving a narrow margin at the top and bottom.

(ii) Hence, or otherwise, find the values of t for which $f\left(\frac{1}{t}\right) = \frac{9}{t}$.

A full-page sheet of white graph paper with a light gray grid. The grid consists of small squares, approximately 10 units wide by 10 units high. There are no margins or other markings on the page.

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Pre-Junior Certificate, 2017 – Higher Level

Mathematics – Paper 1

Time: 2 hours, 30 minutes

